

Bluetooth and Wi-Fi Engineer Mode Instructions

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Revision history

Version	Date	Notes
V5.2	30-Dec-2024	Update the description of Power Level
V5.1	10-Apr-2024	Optimize content description.
V5.0	23-Feb-2024	• Add information about UMW2661. • Update the description about entering engineer mode. • Optimize content description in the document.
V4.0	19-Dec-2022	Add description about UMW2631.
V3.0	26-Nov-2020	Update the template and revise some description.
V2.0	31-Dec-2019	Add the section to describe the Bluetooth and Wi-Fi engineer Mode for UMW2651/2652(E).
V1.0	27-Jul-2017	This is the first official release.

Keyword

Keyword: **Bluetooth®**, Wi-Fi, Engineer Mode

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01

Entering engineer mode

Entering engineer mode

Follow the steps below to enter engineer mode:

1. Make sure to turn off Wi-Fi and **Bluetooth**® in device settings and **wifi/bt scanning** in **Setting > Location > Location services**.
2. Type *****83781***** in the dialer to enter engineer mode, as shown in Figure 1.
3. Navigate to the **CONNECTIVITY** tab, as shown in Figure 2.
4. Select **BT** or **Wifi** to go to the corresponding menu, and perform related operations. For details, see subsequent sections.



Figure 1 Engineer mode



Figure 2 Connectivity tab

02

SC2342B Bluetooth and Wi-Fi engineer mode

SC2342B Bluetooth and Wi-Fi engineer mode



This section contains the following two parts:

- Bluetooth engineer mode

- Bluetooth engineer mode menu
- Bluetooth signaling test
- Bluetooth non-signaling TX test
- Bluetooth non-signaling TX parameters
- Bluetooth non-signaling RX test
- Bluetooth non-signaling RX parameters
- Non-signaling BLE TX screen and parameters
- Non-signaling BLE RX screen and parameters

- Wi-Fi engineer mode

- Wi-Fi engineer mode menu
- Wi-Fi non-signaling TX test
- Wi-Fi non-signaling TX parameters
- Wi-Fi non-signaling RX test
- Wi-Fi non-signaling RX parameters

Bluetooth engineer mode menu

The Bluetooth engineer mode menu consists of the following functions, as shown in Figure 1:

- **BT CLASSIC BQB:** enter the Bluetooth signaling test mode
- **BT_Non-signaling TEST:** enter the Bluetooth non-signaling test mode
- **BT LE BQB:** enter the BLE BQB test mode
- **BT Debug:** conduct debugging

BT Non-signaling TEST consists of the following four tests, as shown in Figure 2:

- **Non-signaling TX**
- **Non-signaling RX**
- **Non-signaling BLE TX**
- **Non-signaling BLE RX**

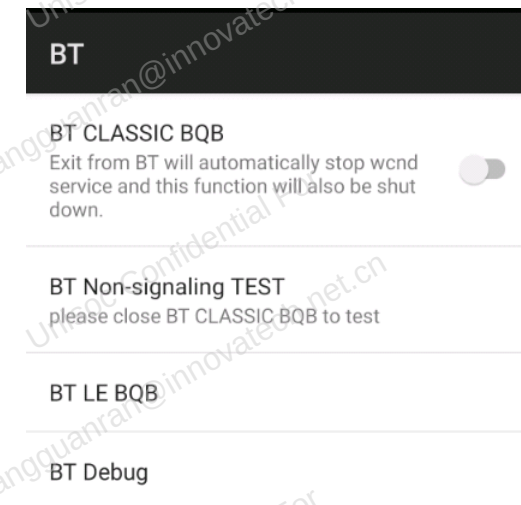


Figure 1

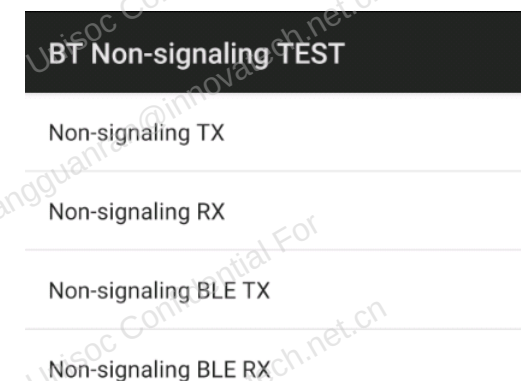


Figure 2

Bluetooth signaling test

The steps of Bluetooth signaling test are as follows:

1. Turn on the **BT CLASSIC BQB** switch in engineer mode of the device to enter the BT EUT mode, as shown in Figure 1.
2. Turn on the **Bluetooth Signaling** switch on the instrument, click **Inquire** > **Connect Test Mode**, as shown in Figure 2. When it shows **CONNECTED TESTMODE** in the **Connection Status** area, which indicates successful connection, you can perform Bluetooth Classic TX/RX signaling test.

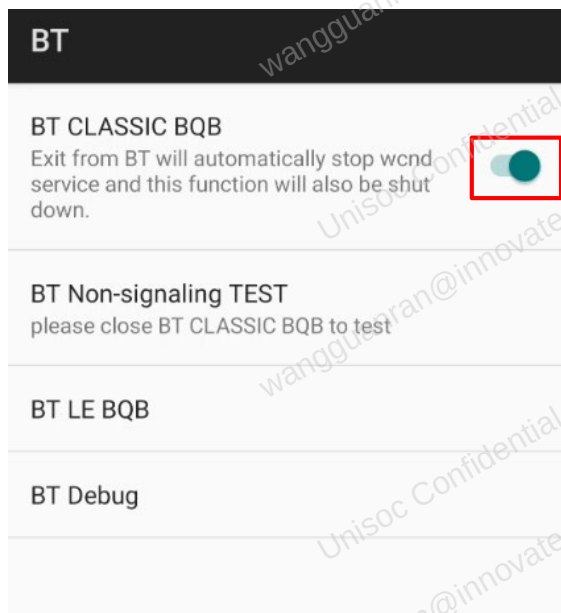


Figure 1 Engineer mode settings

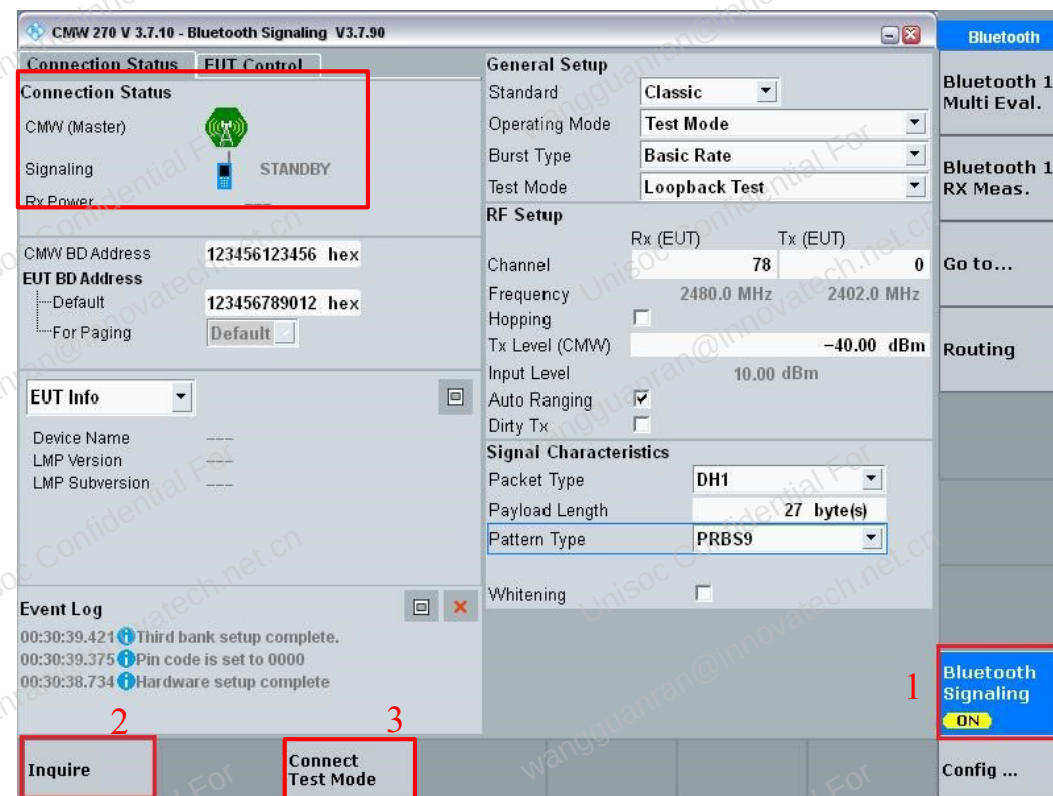


Figure 2 Instrument settings

Bluetooth non-signaling TX test

The steps of Bluetooth non-signaling TX test are as follows:

1. Go to the **Non Signaling TX** screen.
2. Select **CLASSIC** from the **TX Mode** drop-down list.
3. Set other Bluetooth packet parameters.
4. Tap **START**.
5. Measure TX signals with a universal radio communication tester.

Note:

- If you need to modify parameters during test, tap **STOP** first, modify parameters, and then **START** to make the modification take effect.
- For TX parameters, see [Bluetooth non-signaling TX parameters](#).

Non Signaling TX

TX Pattern	00000000
TX Channel	255 or 0~78
TX Pac Type	NULLpkt
TX Pac Len MaxLen is 0	
TX Power Value 0~4	
TX Pac Cnt	0
TX Mode	CLASSIC

STARTSTOP

Bluetooth non-signaling TX parameters

The figure shows the Bluetooth non-signaling TX parameters.

- **TX Pattern:** set a packet data type, including 00000000, 11111111, 10101010, 111100000, and PRBS9.
- **TX Channel:** set a channel (fixed frequency) by entering a decimal number ranging from 0 to 78, or entering 255 that indicates the frequency hopping mode.
- **TX Pac Type:** set a packet type, including DH1, DH3, DH5, 2DH1, 2DH3, 2DH5, 3DH1, 3DH3, and 3DH5.
- **TX Pac Len:** set the packet length by entering a decimal number. After setting the **TX Pac Type** field, the maximum length of the selected packet type is displayed under the **TX Pac Len** field. If the length entered exceeds the maximum length, the software selects the maximum length of the selected packet type by default.
- **TX Power Value:** set TX power level by entering a decimal number. For SC2342B, the value range is 0 to 4. The number 4 indicates the maximum power, and the number 0 the minimum one. Each value corresponds to a value set in power control.
- **TX Pac Cnt:** set the TX packet counts by entering a decimal number ranging from 0 to 65536. The number 0 indicates continuous transmission. The number 12345 indicates sending single tone signals, and the number 56789 indicates sending the single tone signals with frequency offset by 170kHz. It is suggested to set the value to 0.
- **TX Mode:** there are two modes, namely **CLASSIC** and **CW**. Select **CLASSIC** for common test, while **CW** for single carrier test.

Non Signaling TX

TX Pattern	00000000	▼
TX Channel	255 or 0~78	
TX Pac Type	NULLpkt	▼
TX Pac Len	MaxLen is 0	
TX Power Value	0~4	
TX Pac Cnt	0	
TX Mode	CLASSIC	▼
STARTSTOP		

Bluetooth non-signaling RX test



The steps of Bluetooth non-signaling RX test are as follows:

1. Connect the device to the universal radio communication tester.
2. Go to the **Non-signaling RX** screen to set Bluetooth packet parameters.
3. Tap **START**.
4. Send signals from the universal radio communication tester.
5. Read the received data from the test device side. There are two ways to read the received data:
 - Tap **READ** to update data manually.
 - Tap **AUTO** to set the time interval of automatic data update.

RSSI: display the signal strength received at the antenna port.

PER: (packet error count/total packet count) × 100%=PER

BER: (bit error count/total bit count) × 100%=BER

Swipe the screen to the left to read the data if the data length exceeds the screen width.

Note:

- If you need to modify parameters during test, tap **STOP** first, modify parameters, and then **START** to make the modification take effect.
- Make sure there is RF signal input before tapping **READ**.
- Tap **CLEAR** to refresh the screen.
- Tap **STOP** to clear the historical data received.

Non-signaling RX

RX Pattern00000000

RX Channel0~78

RX Pac TypeNULLpkt

RX Gain0~32

RX Addrinput 12 bits addr

RSSI

PER

BER

START

READ

AUTO

CLEAR

STOP

Bluetooth non-signaling RX parameters

The figure shows the Bluetooth non-signaling RX parameters.

- **RX Pattern:** set a packet data type, including 00000000, 11111111, 10101010, 111100000 and PRBS9.
- **RX Channel:** set a channel to receive signals by entering a decimal number ranging from 0 to 78 (2402MHz to 2480MHz).
- **RX Pac Type:** set a packet type, including DH1, DH3, DH5, 2DH1, 2DH3, 2DH5, 3DH1, 3DH3, and 3DH5.
- **RX Gain:** set a RX gain level by entering a decimal number. The number 0 stands for AGC mode, while the numbers from 1 to 32 for fixed gain.
- **RX Addr:** set the Bluetooth address of the universal radio communication tester. It is a 12-bit hexadecimal number by default, for example, 123456123456.

Note: The value of the **RX Addr** field must be consistent with the Bluetooth address of the Bluetooth file in the signal source.

Non-signaling RX

RX Pattern

00000000

RX Channel

0~78

RX Pac Type

NULLpkt

RX Gain

0~32

RX Addr

input 12 bits addr

RSSI

PER

BER

START

READ

AUTO

CLEAR

STOP

Non-signaling BLE TX screen and parameters

The figure shows the non-signaling BLE TX parameters.

- **BLE TX LE_PHY**: select a test rate.
- **BLE TX Pattern**: set a packet data type, including 11111111,10101010, PRBS9, and 11110000.
- **BLE TX Channel**: set a channel (fixed frequency) by entering a decimal number ranging from 0 to 39 (2402MHz ~ 2480MHz) .
- **BLE TX Data Length**: set the data length by entering a decimal number ranging from 0 to192. The maximum length of the selected data type is shown under the **BLE TX Data Length** field.
- **BLE TX Pac Cnt**: set the BLE TX packet counts by entering a decimal number ranging from 0 to 65536. The number 0 indicates continuous transmission. The number 12345 indicates sending single tone signals, and the number 56789 indicates sending the single tone signals with frequency offset by 170kHz. It is suggested to set the value to 0.
- **TX Mode**: There are two modes, namely **BLE** and **CW**. Select **BLE** for common test, while **CW** for single carrier test.

Non-Signaling BLE TX

BLE TX LE_PHY	LE1M	▼
BLE TX Pattern	01010101	▼
BLE TX Channel	0~39	
BLE TX Data Length	0~192	
BLE TX Pac Cnt	0	
TX Mode	BLE	▼

START

STOP

Non-signaling BLE RX screen and parameters

The figure shows the non-signaling BLE RX parameters.

- **BLE RX Channel:** set a channel to receive signals by entering a decimal number ranging from 0 to 39 (2402MHz ~ 2480MHz).
- **BLE RX Gain:** set a RX gain level by entering a decimal number. The number 0 stands for AGC mode, while the numbers from 1 to 5 for the fixed gain levels.
- **BLE RX Addr:** set the Bluetooth address of the universal radio communication tester. It is a 12-bit hexadecimal number by default, for example, 123456123456.

Note: The value of the **BLE RX Addr** field must be consistent with the Bluetooth address of the Bluetooth file in the signal source.

Non Signaling BLE RX

BLE RX Channel 0~39

BLE RX Gain

0~5

BLE RX Addr

input 12-bits addr

RSSI

PER

START

READ

AUTO

CLEAR

STOP

Wi-Fi engineer mode menu



Wi-Fi engineer mode mainly consists of two parts:

- TX (NON SIGNALING TX)
- RX (NON SIGNALING RX)

Note: You can only test Wi-Fi non-signaling TX and RX in engineer mode.
Wi-Fi signaling test requires the connection to the instrument Wi-Fi.

Wifi	
TX	
RX	
REG_WR	
Power Save	☐
power save mode disabled	

Wi-Fi non-signaling TX test

The steps of TX modulating signal test are as follows:

1. Go to the **Wifi TX** screen.
2. Set parameters. For details, see [Wi-Fi non-signaling TX Parameters](#).
3. Tap **START**.
4. Measure TX signals with the universal radio communication tester.

The steps of TX single-carrier signal test are as follows:

1. Go to the **Wifi TX** screen.
2. Set parameters:
 - set **Pkt length**: 1000
 - set **Channel**
 - set **Mode**: Sin Wave
3. Tap **START**.
4. Measure TX signals with the universal radio communication tester.

Note:

If you need to modify parameters during test, tap **STOP** first, modify parameters, and then **START** to make the modification take effect.

Wifi TX

Pkt length	1000
Pkt cnt	0
Power level	0~17
RF Standard	802.11b
Channel	CH1[2412M]
Rate	1Mbps
Preamble	Normal
Mode	Sin Wave
Guard interval	400ns

START**STOP**

Wi-Fi non-signaling TX parameters

- **Pkt length**: set the packet length. It is usually set to 1000.
- **Pkt cnt**: set the TX packet counts. The number 0 stands for continuous transmission.
- **Power level**: set a Wi-Fi TX power level. There is a difference between the power level and the measured power value, and the actual power is subject to the measured power.
- **RF Standard**: 802.11b/g/n_2.4G.
- **Channel**: set the TX channel (fixed frequency), channel 1 to 14.
- **Rate**: set a Wi-Fi rate. Different RF standards correspond to different rate options.
- **Preamble**: the options include **Normal**, **CCK short**, **802.11n Mixed Mode**, **802.11n**.
- **Mode**: the options include **802.11 pkt** and **Sin Wave**.
- **Guard interval**: the options include **400ns** and **800ns**.

Wifi TX	
Pkt length	1000
Pkt cnt	0
Power level	0~17
RF Standard	802.11b
Channel	CH1[2412M]
Rate	1Mbps
Preamble	Normal
Mode	Sin Wave
Guard interval	400ns
STARTSTOP	

Wi-Fi non-signaling RX test

The steps of Wi-Fi non-signaling RX test are as follows:

1. Go to the **Wifi RX** screen.
2. Set parameters. For details, see [Wi-Fi non-signaling RX parameters](#).
3. Tap **START**.
4. Send signals from the universal radio communication tester.
5. Tap **STOP**.

Note:

- After tapping **STOP**, the **RX Ok** field displays the count of the received Wi-Fi packets, and the **PER** field shows the packet error rate.

$$\text{PER} = (\text{packet error count} / \text{total packet count}) \times 100\%$$

- The **STOP** button contains the following commands:

- get_rx_ok
- rx stop

- The value of the **Test RX Num** field should be consistent with the packet number set in signal source.



The screenshot shows the 'Wifi RX' test interface. At the top, there's a title bar 'Wifi RX'. Below it, the 'Test RX Num' is set to 0. The 'RF Standard' is set to '802.11b' and the 'Channel' is set to 'CH1[2412M]'. There are two buttons, 'START' and 'STOP', at the bottom. The 'RX Ok' and 'PER' fields are currently empty.

Wifi RX	
Test RX Num	0
RF Standard	802.11b
Channel	CH1[2412M]
RX Ok:	
PER:	
START STOP	

Wi-Fi non-signaling RX parameters



- **Test RX Num:** set the number of test packets. This value is the same as the number of packets sent by the instrument.
- **RF Standard:** 802.11b/g/n_2.4G.
- **Channel:** set a RX channel (fixed frequency), channel 1 to 14.
- **RX Ok:** the number of received Wi-Fi packets. The ratio of this value to the number of Wi-Fi packets sent by the instrument is used to confirm PER (packet error rate) of the test.
- **PER:** packet error rate.

Wifi RX

Test RX Num 0

RF Standard 802.11b

Channel CH1[2412M]

RX Ok:

PER:

START

STOP

03
**UMW2631/2651/
2652(E)**
**Bluetooth and Wi-Fi
engineer mode**

UMW2631/2651/2652(E) Bluetooth and Wi-Fi engineer mode



This section contains the following two parts:

- Bluetooth engineer mode

- Bluetooth engineer mode menu
- Bluetooth signaling test
- Bluetooth non-signaling TX test
- Bluetooth non-signaling TX parameters
- Bluetooth non-signaling RX test
- Bluetooth non-signaling RX parameters
- Non-Signaling BLE TX screen and parameters
- Non-Signaling BLE RX screen and parameters

- Wi-Fi engineer mode

- Wi-Fi engineer mode menu
- Wi-Fi non-signaling TX test
- Wi-Fi non-signaling TX parameters
- Wi-Fi non-signaling RX test
- Wi-Fi non-signaling RX parameters

Bluetooth engineer mode menu

The Bluetooth engineer mode menu consists of the following functions, as shown in Figure 1:

- **RF Path:** switch the path between **alone** and **shared**. It is shared path when the switch is turned on.
- **BT CLASSIC BQB:** enter the Bluetooth signaling test mode
- **BT_Non-signaling_TEST:** enter the Bluetooth non-signaling test mode
- **BT LE BQB:** enter the BLE BQB test mode
- **BT Debug:** conduct debugging

Note: UMW2652(E) does not have the **RF Path** switch, so you do not need to set the path, but select the test mode directly.

The **BT Non-signaling TEST** screen consist of the following 4 tests, as shown in Figure 2:

- **Non-signaling TX**
- **Non-signaling RX**
- **Non-signaling BLE TX**
- **Non-signaling BLE RX**



Figure 1

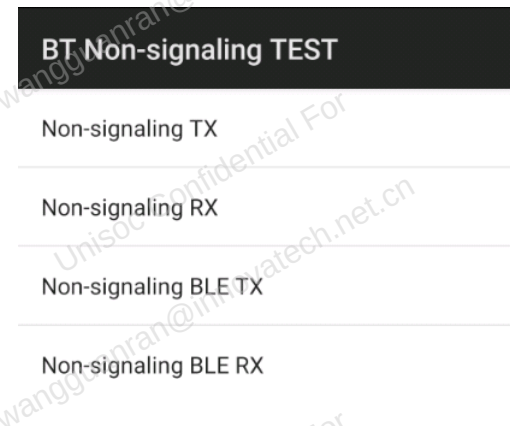


Figure 2

Bluetooth signaling test

The steps of Bluetooth signaling test are as follows:

1. In engineer mode on the device, turn on **RF Path** to switch to the shared path, and turn on the **BT CLASSIC BQB** switch to enter the BT EUT mode, as shown in Figure 1.
2. Turn on the **Bluetooth Signaling** switch on the instrument, click **Inquire** > **Connect Test Mode**, as shown in Figure 2. When it shows **CONNECTED TESTMODE** in the **Connection Status** area, which indicates successful connection, you can perform Bluetooth Classic TX/RX signaling test.

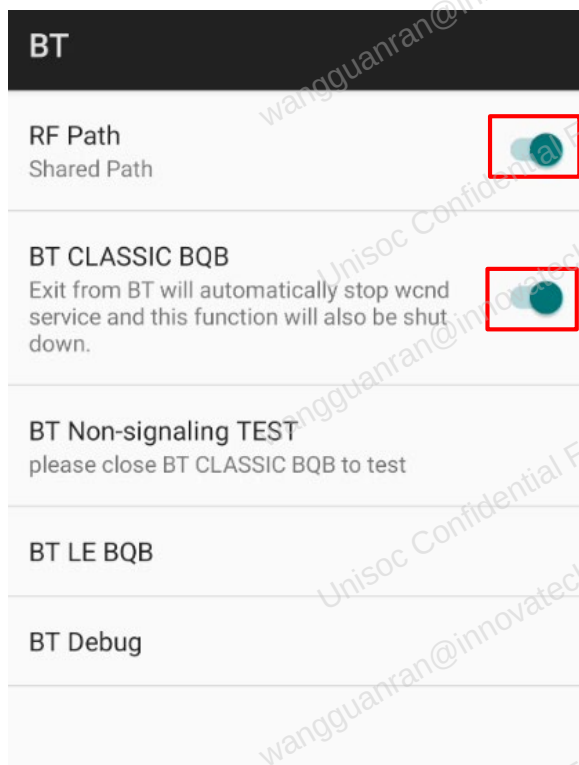


Figure 1 Engineer mode settings

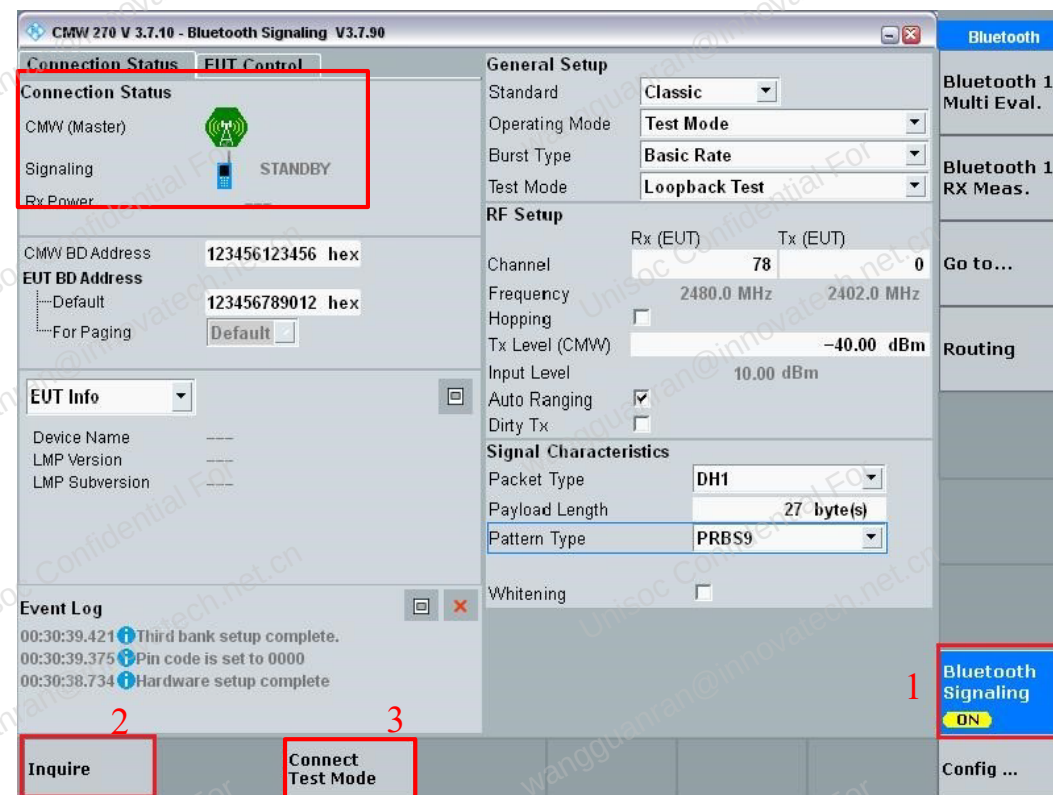


Figure 2 Instrument settings

Bluetooth non-signaling TX test

The steps of Bluetooth non-signaling TX test are as follows:

1. Go to the **Non Signaling TX** screen.
2. Select **CLASSIC** from the **TX Mode** drop-down list.
3. Set other Bluetooth packet parameters.
4. Tap **START**.
5. Measure TX signals with the universal radio communication tester.

Note:

- If you need to modify parameters during test, tap **STOP** first, modify parameters, and then **START** to make the modification take effect.
- For TX parameters, see [Bluetooth non-signaling TX parameters](#).

Non Signaling TX

TX Pattern	00000000	▼
TX Channel	255 or 0~78	
TX Pac Type	NULLpkt	▼
TX Pac Len	MaxLen is 0	
TX Power Value	0~9	
TX Pac Cnt	0	
TX Mode	CLASSIC	▼

START **STOP**

Bluetooth non-signaling TX parameters

The left figure shows the Bluetooth non-signaling TX parameters.

- **TX Pattern:** set a packet data type, including 00000000, 11111111, 10101010, 111100000, and PRBS9.
- **TX Channel:** set a channel (fixed frequency) by entering a decimal number ranging from 0 to 78, or entering 255 that indicates the frequency hopping mode.
- **TX Pac Type:** set a packet type, including DH1, DH3, DH5, 2DH1, 2DH3, 2DH5, 3DH1, 3DH3, and 3DH5.
- **TX Pac Len:** set the packet length by entering a decimal number. After setting the **TX Pac Type** field, the maximum length of the selected packet type is displayed under the **TX Pac Len** field. If the length entered exceeds the maximum length, the software selects the maximum length of the selected packet type by default.
- **TX Power Value:** set a TX power level by entering a decimal number. For UMW2631/2651/2652(E), the value range is 0 ~ 9. The number 9 indicates the maximum power, and the number 0 the minimum power. Each value corresponds to a value set in power control.
- **TX Pac Cnt:** set the TX packet counts by entering a decimal number ranging from 0 to 65536. The number 0 indicates continuous transmission. The number 12345 indicates sending single tone signals, and the number 56789 indicates sending the single tone signals with frequency offset by 170kHz. It is suggested to set the value to 0.
- **TX Mode:** there are two modes, namely **CLASSIC** and **CW**. Select **CLASSIC** for common test, while **CW** for single carrier test.

Non Signaling TX

TX Pattern	00000000	▼
TX Channel	255 or 0~78	
TX Pac Type	NULLpkt	▼
TX Pac Len	MaxLen is 0	
TX Power Value	0~9	
TX Pac Cnt	0	
TX Mode	CLASSIC	▼
START		STOP

Bluetooth non-signaling RX test

The steps of Bluetooth non-signaling RX test are as follows:

1. Connect the device with the universal radio communication tester.
2. Go to the **Non-signaling RX** screen to set Bluetooth packet parameters.
3. Tap **START**.
4. Send signals from the universal radio communication tester.
5. Read the received data from the device side. There are two ways to read the received data:
 - Tap **READ** to update data manually.
 - Tap **AUTO** to set the time interval of automatic data update.

RSSI: display the signal strength received at the antenna port.

PER: (packet error count/total packet count) × 100%=PER

BER: (bit error count/total bit count) × 100%=BER

Swipe the screen to the left to read the data if the data length exceeds the screen width.

Note:

- If you need to modify parameters during test, tap **STOP** first, modify parameters, and then **START** to make the modification take effect.
- Make sure there is RF signal input before tapping **READ**.
- Tap **CLEAR** to refresh the screen.
- Tap **STOP** to clear the historical data received.

Non-signaling RX

RX Pattern	00000000	▼
RX Channel	0~78	
RX Pac Type	NULLpkt	▼
RX Gain	0~32	
RX Addr	input 12 bits addr	
RSSI	PER	BER

START

READ

AUTO

CLEAR

STOP

Bluetooth non-signaling RX parameters

The left figure shows the Bluetooth non-signaling RX parameters.

- **RX Pattern:** set a packet data type, including 00000000, 11111111, 10101010, 111100000 and PRBS9.
- **RX Channel:** set a channel to receive signal by entering a decimal number ranging from 0 to 78 (2402MHz to 2480MHz).
- **RX Pac Type:** set a packet type, including DH1, DH3, DH5, 2DH1, 2DH3, 2DH5, 3DH1, 3DH3, and 3DH5.
- **RX Gain:** set a RX gain level by entering a decimal number. The number 0 stands for AGC mode, while the numbers from 1 to 32 for fixed gain.
- **RX Addr:** set the Bluetooth address of the universal radio communication tester. It is a 12-bit hexadecimal number by default, for example, 123456123456.

Note: The value of the **RX Addr** field must be consistent with the Bluetooth address of the Bluetooth file in the signal source.

Non-signaling RX

RX Pattern

00000000

RX Channel

0~78

RX Pac Type

NULLpkt

RX Gain

0~32

RX Addr

input 12 bits addr

RSSI

PER

BER

START

READ

AUTO

CLEAR

STOP

Non-signaling BLE TX screen and parameters

The figure shows the non-signaling BLE TX parameters.

- **BLE TX LE_PHY**: select a test rate.
- **BLE TX Pattern**: set a packet data type, including 11111111,10101010, PRBS9, and 11110000.
- **BLE TX Channel**: set a channel (fixed frequency) by entering a decimal number ranging from 0 to 39 (2402MHz to 2480MHz) .
- **BLE TX Data Length**: set a data length by entering a decimal number ranging from 0 to 192. The maximum length of the selected data type is shown under the **BLE TX Data Length** field.
- **BLE TX Pac Cnt**: set the BLE TX packet counts by entering a decimal number ranging from 0 to 65536. The number 0 indicates continuous transmission. The number 12345 indicates sending single tone signals, and the number 56789 indicates sending the single tone signals with frequency offset by 170kHz. It is suggested to set the value to 0.
- **Tx Mode**: there are two modes, namely **BLE** and **CW**. Select **BLE** for common test, while **CW** for single carrier test.

Non-Signaling BLE TX

BLE TX LE_PHY	LE1M	▼
BLE TX Pattern	01010101	▼
BLE TX Channel	0~39	
BLE TX Data Length	0~192	
BLE TX Pac Cnt	0	
TX Mode	BLE	▼
STARTSTOP		

Non-signaling BLE RX screen and parameters

The figure shows the non-signaling BLE RX parameters.

- **BLE RX Mod_Index:** select a modulation index. Currently only **Standard** is supported.
- **BLE RX LE_PHY:** select a test rate.
- **BLE RX Channel:** set a channel to receive signals by entering a decimal number ranging from 0 to 39 (2402MHz to 2480MHz).
- **BLE RX Gain:** set a RX gain level by entering a decimal number. The number 0 stands for AGC mode, while the numbers from 1 to 5 for the fixed gain levels.

Non Signaling BLE RX	
BLE RX Mod_Index	Standard
BLE RX LE_PHY	LE1M
BLE RX Channel	0~39
BLE RX Gain	0~5
RSSI	PER

STARTREADAUTOCLEARSTOP

Wi-Fi engineer mode menu



Wi-Fi engineer mode mainly consists of two parts:

- TX (NON SIGNALING TX)
- RX (NON SIGNALING RX)

Note: You can only test Wi-Fi non-signaling TX and RX in engineer mode.
Wi-Fi signaling test requires the connection to the instrument Wi-Fi.

Wifi	
TX	
RX	
REG_WR	
Power Save	☐
power save mode disabled	

Wi-Fi non-signaling TX test

The steps of TX modulating signal test are as follows:

1. Go to the **Wifi TX** screen.
2. Set parameters. For details, see [Wi-Fi non-signaling TX parameters](#).
3. Tap **START**.
4. Measure TX signals with the universal radio communication tester.

The steps of TX single-carrier signal test are as follows:

1. Go to the **Wifi TX** screen.
2. Set parameters:
 - set **Pkt length**: 1000
 - set **Channel**
 - set **Mode**: Sin wave
3. Tap **START**.
4. Measure TX signals with the universal radio communication tester.

Note:

- If you need to modify parameters during test, tap **STOP** first, modify parameters, and then **START** to make the modification take effect.
- UMW2652(E) does not have the **RF Tx Path** field on the **Wifi TX** screen.

2:15 AM

Wifi TX

Pkt length	64~4095
Pkt cnt	0~65535
Power level	0~24 ☐
RF Standard	802.11b ▼
RF Tx Path	Primary ▼
CBW	20MHz ▼
SBW	20MHz ▼
Offset	0MHz ▼
Channel	CH1[2412M] ▼
Rate	1M_Long ▼
Preamble	Normal ▼
Mode	802.11 pkt ▼
Guard interval	400ns ▼

START **STOP**

Wi-Fi non-signaling TX parameters

- **Pkt length:** set a packet length. It is usually set to 1000.
- **Pkt cnt:** set the TX packet counts. The number 0 stands for continuous transmission.
- **Power level:** set a Wi-Fi TX power level. If the check box on the right is not selected, the power configured in ini is used by default. If the check box is selected, input the required power level. Please be noted there is a difference between the power level and the measured power value, and the actual power is subject to the measured power.
- **RF Standard:** 802.11b/g/n_2.4G/n_5.0G/ac/a.
- **RF Tx Path:** select **Primary**, **Diversity** or **MIMO** antenna.
- **CBW:** set the channel band width.
- **SBW:** set the signal band width.
- **Offset:** set the frequency offset.
- **Channel:** set the TX channel (2.4G/5G fixed frequency).
- **Rate:** set a Wi-Fi rate. Different RF standards correspond to different rate options.
- **Preamble:** the options include **Normal**, **CCK short**, **802.11n Mixed Mode**, and **802.11n**. **Normal** is selected by default.
- **Mode:** select to send modulated wave or single carrier wave. The options include **802.11 pkt** and **Sin Wave**.
- **Guard interval:** the options include **400ns** and **800ns**.

2:15 AM

Wifi TX

Pkt length	64~4095	
Pkt cnt	0~65535	
Power level	0~24	☐
RF Standard	802.11b	▼
RF Tx Path	Primary	▼
CBW	20MHz	▼
SBW	20MHz	▼
Offset	0MHz	▼
Channel	CH1[2412M]	▼
Rate	1M_Long	▼
Preamble	Normal	▼
Mode	802.11 pkt	▼
Guard interval	400ns	▼

START **STOP**

Wi-Fi non-signaling RX test

The steps of Wi-Fi non-signaling RX test are as follows:

1. Go to the **Wifi RX** screen.
2. Set parameters. For details, see [Wi-Fi non-signaling RX parameters](#).
3. Tap **START**.
4. Send signals from the universal radio communication tester.
5. Tap **STOP**.

Note:

- After tapping **STOP**, the **RX Ok** field displays the count of the received Wi-Fi packets, and **PER** shows the packet error rate.

$$\text{PER} = (\text{packet error count} / \text{total packet count}) \times 100\%$$

- The **STOP** button contains the following commands:
 - get_rx_ok
 - rx stop
- The value of the **Test RX Num** field should be consistent with the packet number set in signal source.
- UMW2652(E) does not have the **RF Rx Path** field on the **Wifi RX** screen.



The screenshot shows the 'Wifi RX' configuration interface. At the top, there's a title bar 'Wifi RX'. Below it, the 'Test RX Num' is set to '0'. A list of parameters follows, each with a dropdown arrow: 'RF Standard' is '802.11b', 'RF Rx Path' is 'Primary', 'CBW' is '20MHz', 'SBW' is '20MHz', 'Offset' is '0MHz', and 'Channel' is 'CH1[2412M]'. Below these, there are fields for 'RX Ok:' and 'PER:'. At the bottom, there are two buttons: 'START' and 'STOP'.

Wifi RX	
Test RX Num	0
RF Standard	802.11b
RF Rx Path	Primary
CBW	20MHz
SBW	20MHz
Offset	0MHz
Channel	CH1[2412M]
RX Ok:	
PER:	
START STOP	

Wi-Fi non-signaling RX parameters

- **Test RX Num:** set the number of test packets. This value is the same as the number of packets sent by the instrument.
- **RF Standard:** 802.11b/g/n_2.4G/n_5.0G/ac/a.
- **RF Rx Path:** select **Primary**, **Diversity** or **MIMO** antenna.
- **CBW:** set the channel band width.
- **SBW:** set the signal band width.
- **Offset:** set the frequency offset.
- **Channel:** set the RX channel (2.4G/5G fixed frequency).
- **RX Ok:** the number of received Wi-Fi packets. The ratio of this value to the number of packets sent by the instrument is used to confirm PER (packet error rate) of the test.
- **PER:** packet error rate.

Wifi RX

Test RX Num 0

RF Standard

802.11b

▼

RF Rx Path

Primary

▼

CBW

20MHz

▼

SBW

20MHz

▼

Offset

0MHz

▼

Channel

CH1[2412M]

▼

RX Ok:

PER:

START

STOP

04

UMW2661 Wi-Fi engineer mode

UMW2661 Wi-Fi engineer mode



This section contains the following parts:

- Wi-Fi engineer mode menu
- Wi-Fi non-signaling TX test
- Wi-Fi non-signaling TX parameters
- Wi-Fi non-signaling RX test
- Wi-Fi non-signaling RX parameters
- DBDC Mode non-signaling test
- DBDC Mode non-signaling parameters

Wi-Fi engineer mode menu

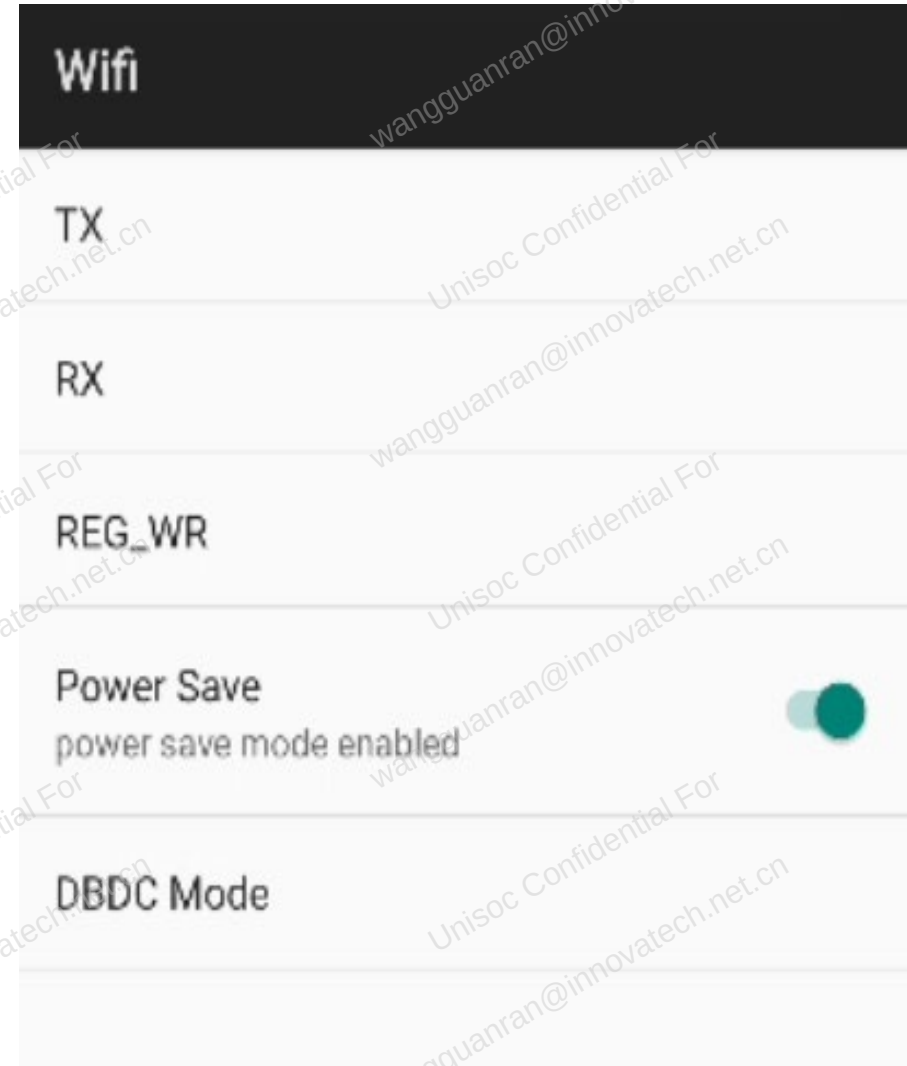


Wi-Fi engineer mode mainly consists of three parts:

- TX (NON SIGNALING TX)
- RX (NON SIGNALING RX)
- DBDC Mode

Note:

- Wi-Fi signaling test does not require entering engineer mode, but only turning on the Wi-Fi switch to connect with the instrument Wi-Fi for testing directly.
- Keep the default setting for the **Power Save** switch.



Wi-Fi non-signaling TX test

The steps of TX modulating signal test are as follows:

1. Go to the **Wifi TX** screen.
2. Set parameters. For details, see [Wi-Fi non-signaling TX parameters](#).
3. Tap **START**.
4. Measure TX signals with the universal radio communication tester.

The steps of TX single-carrier signal test are as follows:

1. Go to the **Wifi TX** screen.
2. Set the following parameters. For details, see [Wi-Fi non-signaling TX Parameters](#).
 - set **Pkt length**
 - set **Channel**
 - set **Mode**
3. Tap **START**.
4. Measure TX signals with the universal radio communication tester.

Note:

If you need to modify parameters during test, tap **STOP** first, modify parameters, and then **START** to make the modification take effect.

The screenshot shows the 'Wifi TX' configuration interface. It includes a title bar 'Wifi TX' and a list of parameters with their current values and dropdown arrows. At the bottom, there are two buttons: 'START' and 'STOP'.

Parameter	Value	Action
Pkt length	64~4095	
Pkt cnt	0~65535	
Power level	0~24	☐
RF Standard	802.11b	▼
RF Tx Path	Primary	▼
CBW	20MHz	▼
SBW	20MHz	▼
Offset	0MHz	▼
Channel	CH1[2412M]	▼
Rate	1M_Long	▼
Preamble	Normal	▼
Mode	802.11 pkt	▼

START **STOP**

Wi-Fi non-signaling TX parameters 1/2

- **Pkt length:** set a packet length. The value ranges from 64 to 4095, and it is usually set to 1000.
- **Pkt cnt:** set the TX packet counts. The value ranges from 0 to 65535. The number 0 stands for continuous transmission.
- **Power level:** set a Wi-Fi TX power level. If the check box on the right is not selected, the power configured in ini is used by default. If the check box is selected, input the required power level. Please be noted there is a difference between the power level and the measured power value, and the actual power is subject to the measured power.
- **RF Standard:** 802.11b/g/n_2.4G/n_5.0G/ac/a/ax_2.4G/ax_5G.
- **RF Tx Path:** select **Primary**, **Diversity** or **MIMO** antenna.
- **CBW:** set the channel band width.
- **SBW:** set the signal band width.
- **Offset:** set the frequency offset.

Wifi TX

Pkt length	64~4095	
Pkt cnt	0~65535	
Power level	0~24	☐
RF Standard	802.11b	▼
RF Tx Path	Primary	▼
CBW	20MHz	▼
SBW	20MHz	▼
Offset	0MHz	▼
Channel	CH1[2412M]	▼
Rate	1M_Long	▼
Preamble	Normal	▼
Mode	802.11 pkt	▼

START**STOP**

Wi-Fi non-signaling TX parameters 2/2

- **Channel:** set the TX channel (2.4G/5G fixed frequency).
- **Rate:** set a Wi-Fi rate. Different RF standards correspond to different rate options.
- **Preamble:** the options include **Normal**, **CCK short**, **802.11n Mixed Mode**, **802.11ac**, **HE_SU** and **HE_TB** (not supported by UMW2661 for now).
- **Mode:** select to send modulated wave, single carrier wave or 100% duty cycle. The options include **802.11 pkt**, **Sin Wave** and **100% duty cycle**.
- **Guard interval:** the options include **800ns**, **VHT/HT long GI**, **VHT/HT short GI**, **1xLTF_0.8μs GI**, **2xLTF_0.8μs GI**, **2xLTF_1.6μs GI**, **4xLTF_0.8μs GI**, **4xLTF_3.2μs GI**.
- **FEC:** the options include **Coding type** and **BCC/LDPC**.

Wifi TX

Pkt length	64~4095	
Pkt cnt	0~65535	
Power level	0~24	☐
RF Standard	802.11b	▼
RF Tx Path	Primary	▼
CBW	20MHz	▼
SBW	20MHz	▼
Offset	0MHz	▼
Channel	CH1[2412M]	▼
Rate	1M_Long	▼
Preamble	Normal	▼
Mode	802.11 pkt	▼

START

STOP

Wi-Fi non-signaling RX test

The steps of Wi-Fi non-signaling RX test are as follows:

1. Go to the **Wifi RX** screen.
2. Set parameters. For details, see [Wi-Fi non-signaling RX parameters](#).
3. Tap **START**.
4. Send signals from the universal radio communication tester.
5. Tap **STOP**.

Note:

- After tapping **STOP**, the **RX Ok** field displays the count of the received Wi-Fi packets, and PER shows the packet error rate.
$$\text{PER} = (\text{packet error count} / \text{total packet count}) \times 100\%$$
- RSSI indicates the strength of the received signal. This field can display the data of two paths for MIMO.
- The **STOP** button contains the following commands:
 - get_rx_ok
 - rx stop
- The value of the **Test RX Num** field should be consistent with the packet number set in signal source.

Wifi RX

Test RX Num 0

RF Standard 802.11b

RF Rx Path Primary

CBW 20MHz

SBW 20MHz

Offset 0MHz

Channel CH1[2412M]

RX Ok:

PER:

RSSI:

START STOP

Wi-Fi non-signaling RX parameters

- **Test RX Num:** set the number of test packets. This value is the same as the number of packets sent by the instrument.
- **RF Standard:** 802.11b/g/n_2.4G/n_5.0G/ac/a/ax_2.4G/ax_5G.
- **RF Rx Path:** select **Primary**, **Diversity** or **MIMO** antenna.
- **CBW:** set the channel band width.
- **SBW:** set the signal band width.
- **Offset:** set the frequency offset.
- **Channel:** set the RX channel (2.4G/5G fixed frequency).
- **RX Ok:** the number of received Wi-Fi packets. The ratio of this value to the number of packets sent by the instrument is used to confirm PER (packet error rate) of the test.
- **PER:** packet error rate.
- **RSSI:** the strength of the received signal, which can display the data of two paths for MIMO.

Wifi RX

Test RX Num 0

RF Standard

802.11b

RF Rx Path

Primary

CBW

20MHz

SBW

20MHz

Offset

0MHz

Channel

CH1[2412M]

RX Ok:

PER:

RSSI:

START

STOP

DBDC Mode non-signaling test

DBDC Mode consists of the following parts:

- TX BAND0
- RX BAND0
- TX BAND1
- RX BAND1

The steps of DBDC Mode test are as follows:

1. Set parameters. For details, see the following pages:
 - [Wi-Fi non-signaling TX Parameters](#)
 - [Wi-Fi non-signaling RX Parameters](#)
1. Tap **START**.
2. Measure TX or RX signals with the universal radio communication tester.
3. Tap **STOP**.

Note:

- If you need to modify parameters during test, tap **STOP** first, modify parameters, and then **START** to make the modification take effect.
- You can set different combinations of 2.4G/5G SISO/MIMO TX/RX and test two bands simultaneously.

Wifi DBDC Mode

TX BAND0	RX BAND0	TX BAND1	RX BAND1
Pkt length	64~4095		
Pkt cnt	0~65535		
Power level	0~24 ☐		
RF Standard	802.11b		
RF Tx Path	Primary		
CBW	20MHz		
SBW	20MHz		
Offset	0MHz		
Channel	CH1[2412M]		
Rate	1M_Long		
Preamble	Normal		
Mode	802.11 pkt		
START		STOP	

DBDC Mode non-signaling parameters



The TX BAND0 screen and the TX BAND1 screen are consistent with the Wi-Fi TX screen.

The RX BAND0 screen and the RX BAND1 screen are consistent with the Wi-Fi RX screen.

For detailed parameter description, see the following pages:

- [Wi-Fi non-signaling TX Parameters](#)
- [Wi-Fi non-signaling RX Parameters](#)

Wifi DBDC Mode

TX BAND0

RX BAND0

TX BAND1

RX BAND1

Pkt length

64~4095

Pkt cnt

0~65535

Power level

0~24

☐

RF Standard

802.11b

▼

RF Tx Path

Primary

▼

CBW

20MHz

▼

SBW

20MHz

▼

Offset

0MHz

▼

Channel

CH1[2412M]

▼

Rate

1M_Long

▼

Preamble

Normal

▼

Mode

802.11 pkt

▼

START

STOP

Thank You



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